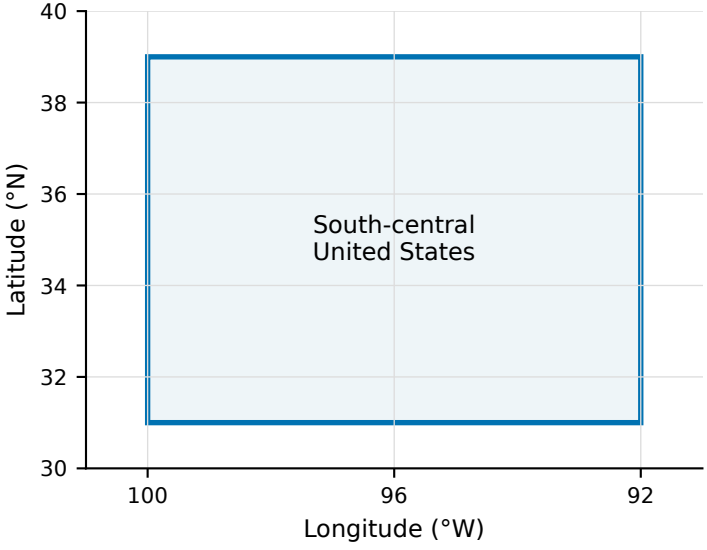
(a) Fixed verification domain



(b) Neighborhoods and endpoints

H: [t − 30 min, t]
Y: (t, t + 30 min]

History includes t; target excludes t

(c) Four verification classes

	Y = 0	Y = 1
H = 0	I− Local-onset negative	I+ Local-onset positive
H = 1	O− Ongoing negative	O+ Ongoing positive

(d) Independent periods

Development: June–July 2024
GOES-16 / MRMS
Training → validation → test

Confirmation: June–July 2025
GOES-19 / MRMS
All 61 dates; frozen rule

Year, weather, and platform are confounded.